Department of Computer Science and Engineering

KPR Institute of Engineering and Technology

LABORATORY RECORD

U21CS503

MOBILE APPLICATION DEVELOPMENTLABORATORY

KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

(Autonomous)

COIMBATORE – 641 407

LABORATORY RECORD

Name : .....

Roll Number : .....

Subject Code & Title: .....

Department : .....

Year & Semester : .....

This is the certified record of work done by.....

Register Number .....

Staff In- Charge Head of the Department

Place:

Date:

He/ She has submitted the record for the End Semester Practical Examination held on .....

Internal Examiner

VISION AND MISSION OF THE INSTITUTION

Vision

To become a premier institute of academic excellence by imparting technical, intellectual and professional skills to students for meeting the diverse needs of the industry, society, the nation and the world at large.

Mission

- . Commitment to offer value-based education and enhancement of practical skills.
- Continuous assessment of teaching and learning process through scholarly activities.
- Enriching research and innovative activities in collaboration with industry and institute of repute.
- Ensuring the academic process to uphold culture,ethicsand social responsibilities

VISION AND MISSION OF THE DEPARTMENT

Vision

To be a premier centre for education, dissemination of knowledge and research in the frontier areas of computer science and engineering to serve the communitywith moral values.

Mission

- Provide holistic education incorporating the state-of-the-art technologiesto produce successful professionals.
- Facilitate the students to pursue higher education and research in theareas related to Computer Science and Engineering.
- Promote strong collaborations with the industries and steer the students to nurture their interest in continuous learning to meet the changing needsof the society.

Program Educational Objectives (PEOs)

Graduates of B.E. CSE program, four years after graduation, will

- PEO1: Acquire knowledge and skills on cutting edge technologies in the field of computer science.
-
- PEO2: Inculcate a passion for continuous learning through further studies and research in the field of computer science and engineering.
-
- PEO3: Develop innovative ideas upholding the rich value systems to solve the changing technological needs.

ANNEXURE – I
THEORY COURSES WITH LAB COMPONENT

Criteria		Excellent (4 Marks)	Good (3 Marks)	Adequate (2 Marks)	Inadequate (1 Mark)
A. Preparation & Observation	**Criterion #1** Ability to setup and conduct experiments.	Able to develop contingency or alternative plans and anticipate problems during experiment.	Able to develop contingency or alternative plans.	Able to use theoretical framework, measurement techniques, testing apparatus or model.	Unable to identify theoretical framework, measurement techniques, testing apparatus or model.
	Criterion #2 Ability to take measurements / readings andpresent data	Able to formulate controls and evaluatealternatives of the experiment . Able to evaluate data and relate to engineering phenomena for decision-making.	Able to evaluate data and relate to engineering phenomena for decision-making.	Able to apply constraint and assumption into the experimental design. Able to conduct experiment correctly and collect data.	Unable to discuss experimental processesand protocols
B. Results & Interpretation	**Criterion #3** Ability to analyze the data theoretically and logically to conclude experimental results	Able to combine / organize more than one set of data, interpret data and make meaningful conclusion.	Able to evaluate or compare data and make meaningful conclusion	Able to select and useand apply appropriate techniques or methods to analyze the data.	Unable to select and describe the techniquesor methods of analyzing the data.
	Criterion #4 Ability to interpret and discuss any discrepancies between theoretical and experimental results	Able to verify and/ orvalidate several sets of data and relates toengineering phenomena for decision making.	Able to verify and/ or validate data and relate to engineering phenomena for decision making.	Able to identify and verify how results relate/differ from theory or previous results	Unable to identify howresults relate/ differ from theory or previous results .
C. Viva Voce	**Criterion #5** Demonstrate the ability to effectively respond to questions	Able to listen carefully and respond to questions appropriately; is ableto explain and interpret results to the teacher	Able to listen carefully and respond to questions appropriately	Misunderstand the questions and does not respond appropriately to the teacher, or has some trouble in answering questions	Unable to listen carefully to questions and does not provide an appropriate answer, or is unable to answer questions

LIST OF EXPERIMENTS

S.NO	DATE	NAME OF THE EXPERIMENTS	PAGE NO	MARKS	SIGN
01		Android application for 'Hello World' textview			
02		Android application for user login			
03		Android application for calculator			
04		Android application for show the current location of the user			
05		Android application for user registration and login with local data storage			
06		Android application for BMI calculator			
07		Android application for expense manager			
08		Android application for converting kilometers to miles			
09		Android application for cloud student database			
10		Android application for day-to-day task (to-do) management			

TOTAL		

EX.NO : 01	'Hello, World' application
DATE :	

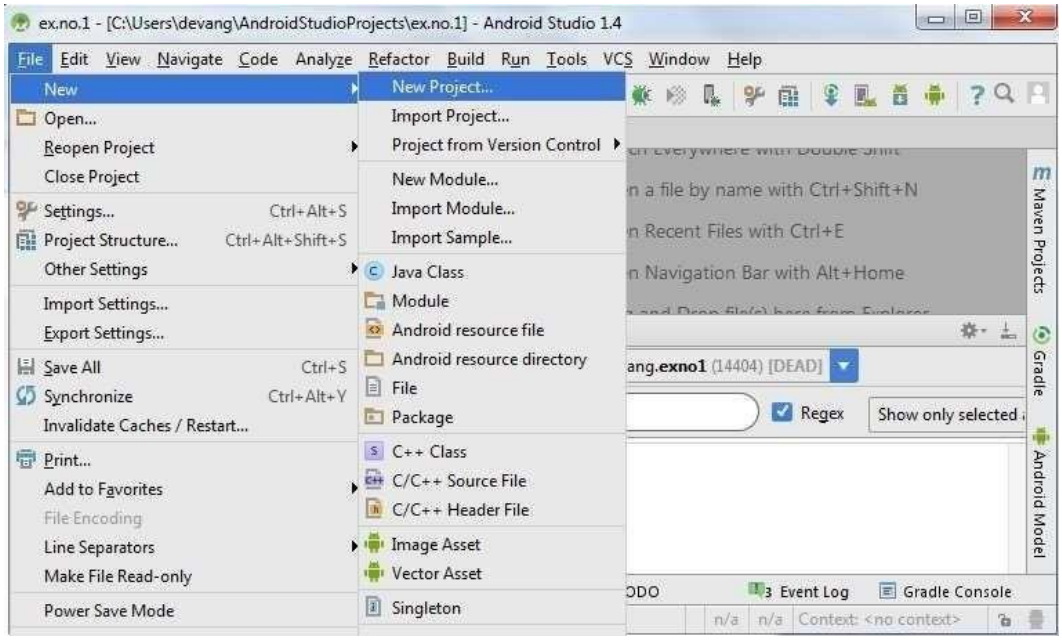
AIM :

To design and developdemontrate a simple android application that display “Hello World”.

Procedure:

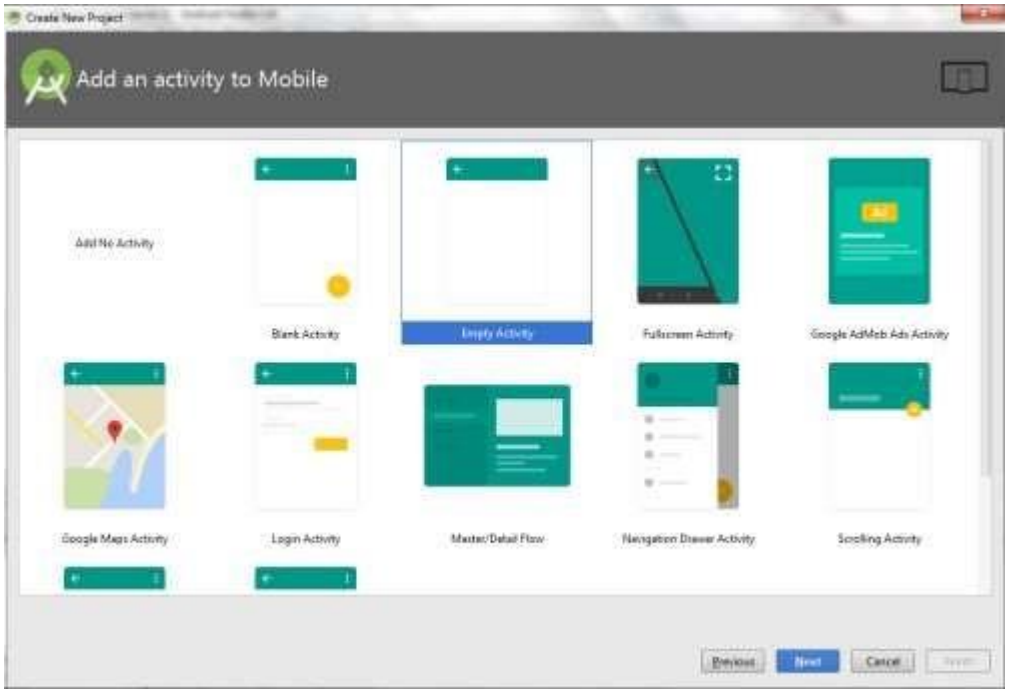
Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.

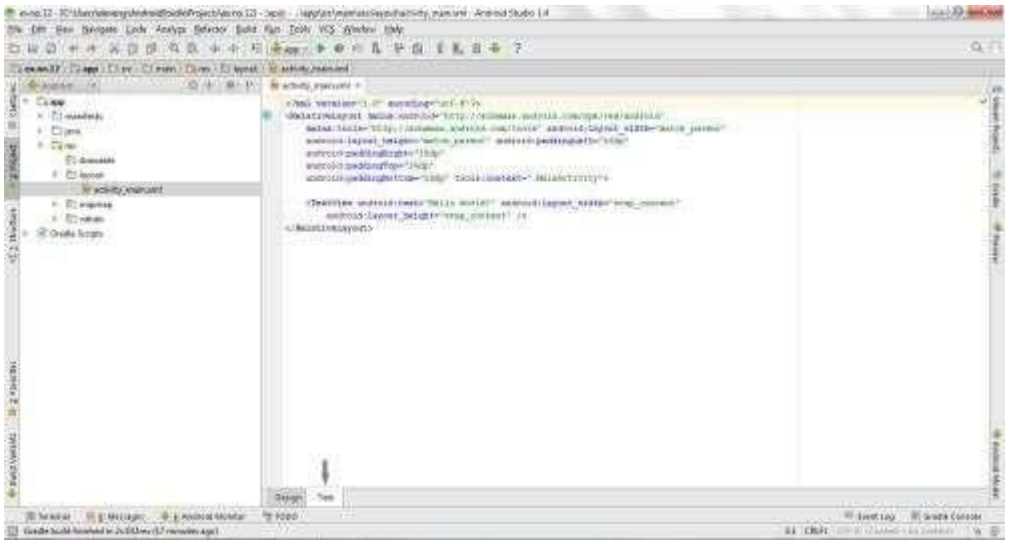


- Then type the Application name as **“Exno1”**and click Next.
- Then **select the Minimum SDK** and click Next.
- Then select the **Empty Activity** and click Next.

.



- Finally click **Finish**.
- It will take some time to load and build



- After completion it will look as given below.

OUTPUT:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

RESULT :

Thus a simple android application that uses text view “Hello World” is developed and executed successfully.

EX.NO : 02	Android application for user login
DATE :	

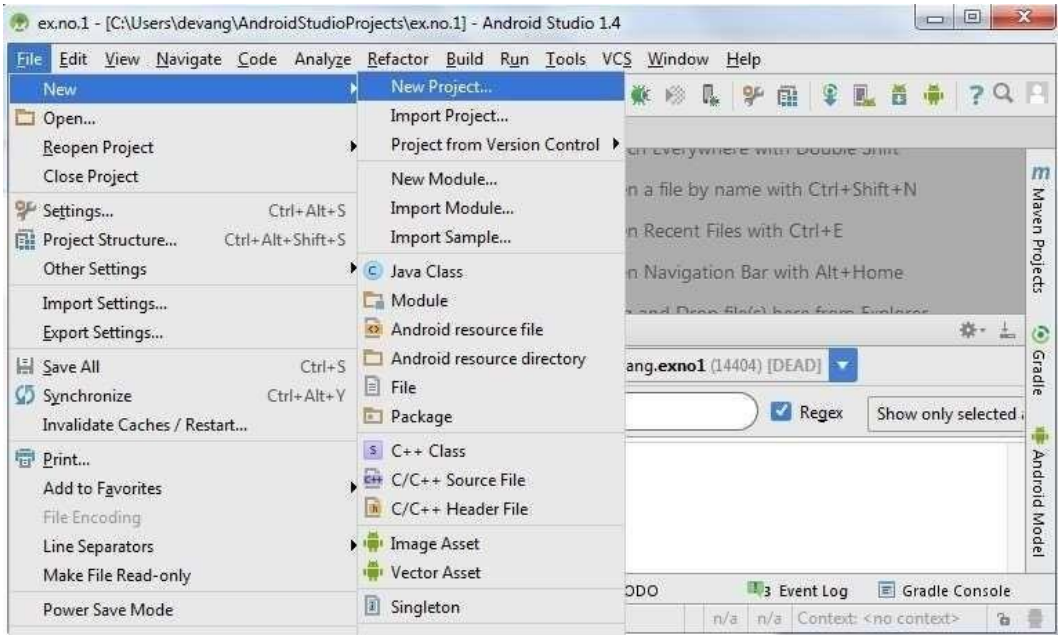
Aim:

To Design an android native application for a user login screen with username, password, reset button and a submit button. Also, include header image and a label. Use layout managers.

Procedure

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.

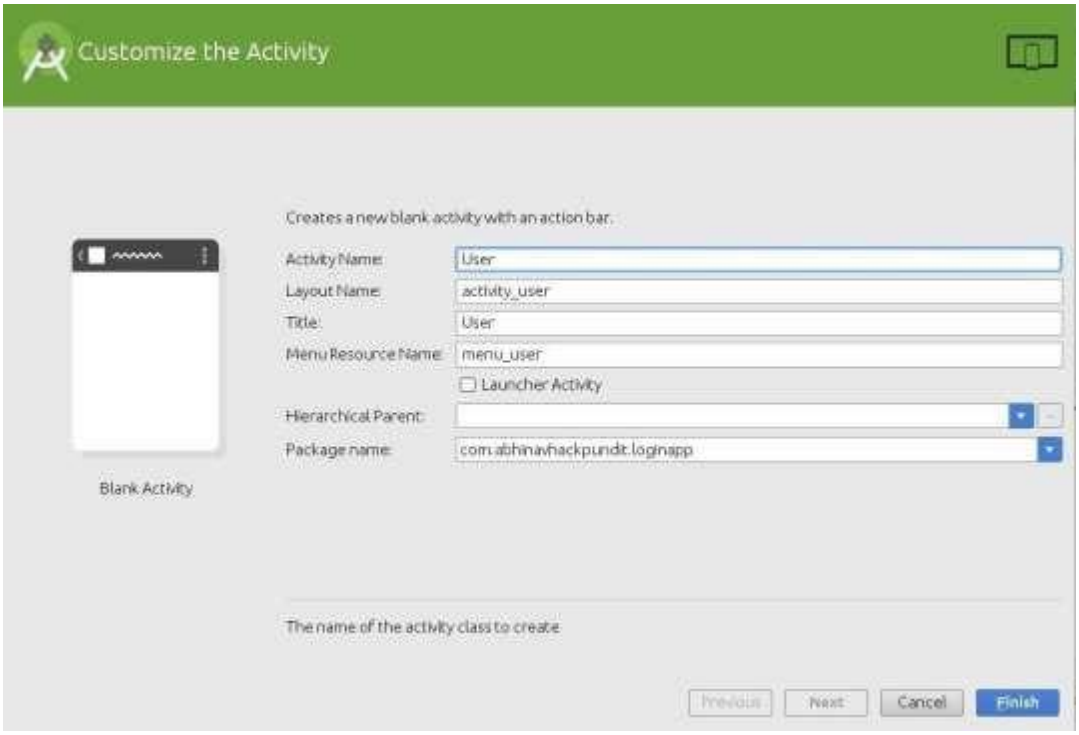


In this Android Simple Login app there are two activities:

- The First activity is to design the **front end**, which is **user view**. Means how user willview our app.
- The Second activity is how to write the **back end code** for the Android Simple LoginApplication.

So lets gets started with our basic Android Simple Login App:

- Open your Android Studio.
- Click on Start a New Android Studio Project.
- Write your **Application Name "LoginApp"** and leave other fields blank as it is
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**.
- Select the **empty activity** and click NEXT.
- Write the activity name "**Login"** and leave everything as it is. Click Finish.



Program:

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
  <ImageView
        android:id="@+id/headerImage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:src="@drawable/header_image"
        app:layout_constraintTop_toTopOf="parent"
```

```
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintBottom_toTopOf="@+id/usernameEditText" />

<EditText
    android:id="@+id/usernameEditText"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Username"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/headerImage"
    app:layout_constraintVertical_chainStyle="packed" />

<EditText
    android:id="@+id/passwordEditText"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/usernameEditText" />
<Button
    android:id="@+id/resetButton"
    android:layout_width="0dp"
```

```
    android:layout_height="wrap_content"
    android:text="Reset"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/passwordEditText"  />
  <Button
    android:id="@+id/submitButton"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Submit"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/resetButton"  />

  <TextView
    android:id="@+id/infoLabel"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Please enter your credentials"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/submitButton"  />
</androidx.constraintlayout.widget.ConstraintLayout>
```

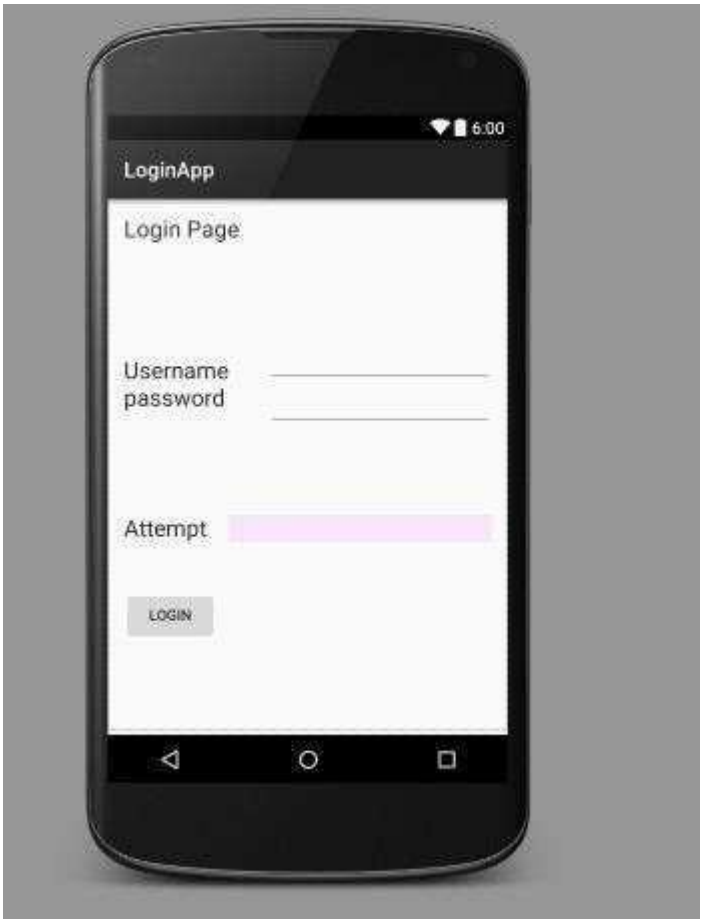
Main Activity.java

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus a simple android user login application is developed and executed successfully.

EX.NO : 03	Android application for calculator
DATE :	

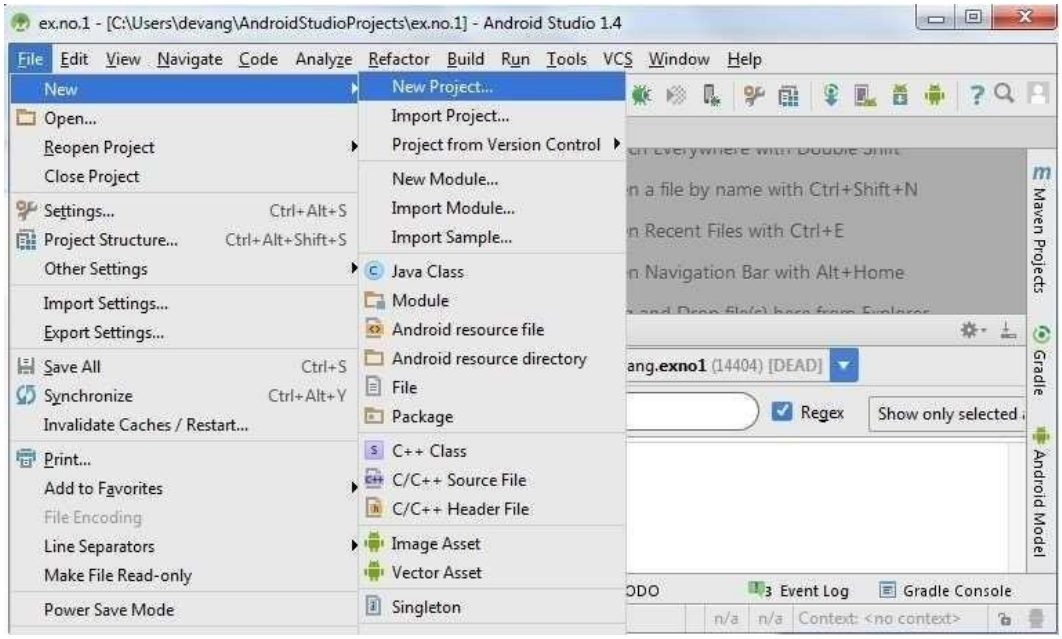
Aim:

To develop a calculator android application with basic arithmetic operation.

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



- Click on Start a New Android Studio Project.
- Write your Application Name **"Calculator"** and leave other fields blank as it is, thenclick NEXT.
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**. After selectingyour API click on NEXT.
- Select the **empty activity** and click NEXT.
- Leave the activity name "**MainActivity"** and leave everything as it is. Click Finish.

Program:

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#8BC34A"
    android:backgroundTint="@android:color/darker_gray"
    tools:context=".MainActivity">

<TextView
    android:layout_width="194dp"
    android:layout_height="43dp"
    android:layout_marginStart="114dp"
    android:layout_marginLeft="114dp"
    android:layout_marginTop="58dp"
    android:layout_marginEnd="103dp"
    android:layout_marginRight="103dp"
    android:layout_marginBottom="502dp"
    android:scrollbarSize="30dp" android:text="
    Calculator"
    android:textAppearance="@style/TextAppearance.AppCompat.Body1"
    android:textSize="30dp" app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/num1" android:layout_width="364dp"
    android:layout_height="28dp"
    android:layout_marginStart="72dp"
    android:layout_marginTop="70dp"
    android:layout_marginEnd="71dp"
    android:layout_marginBottom="416dp"
    android:background="@android:color/white"
    android:ems="10" android:onClick="clearTextNum1"
    android:inputType="number"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/num2" android:layout_width="363dp"
    android:layout_height="30dp"
    android:layout_marginStart="72dp"
    android:layout_marginTop="112dp"
    android:layout_marginEnd="71dp"
```

```
        android:layout_marginBottom="374dp"
        android:background="@android:color/white"
        android:ems="10" android:onClick="clearTextNum2"
        android:inputType="number"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<TextView
        android:id="@+id/result" android:layout_width="356dp"
        android:layout_height="71dp"
        android:layout_marginStart="41dp"
        android:layout_marginTop="151dp"
        android:layout_marginEnd="48dp"
        android:layout_marginBottom="287dp"
        android:background="@android:color/white"
        android:text="result" android:textColorLink="#673AB7"
        android:textSize="25sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

 <Button
        android:id="@+id/sum" android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginTop="292dp"
        android:layout_marginEnd="307dp"
        android:layout_marginBottom="263dp"
        android:backgroundTint="@android:color/holo_red_light"
        android:onClick="doSum"
        android:text="+"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
 <Button
        android:id="@+id/sub"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="210dp"
        android:layout_marginTop="292dp"
```

```
        android:layout_marginEnd="113dp" android:layout_marginBottom="263dp"
        android:backgroundTint="@android:color/holo_red_light"
        android:onClick="doSub"
        android:text="-"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.507" />

<Button
        android:id="@+id/div" android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="307dp"
        android:layout_marginTop="292dp"
        android:layout_marginEnd="16dp"
        android:layout_marginBottom="263dp"
        android:backgroundTint="@android:color/holo_red_light"
        android:onClick="doDiv"
        android:text="/"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

 <Button
        android:id="@+id/mul" android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginTop="356dp"
        android:layout_marginEnd="307dp"
        android:layout_marginBottom="199dp"
        android:backgroundTint="@android:color/holo_red_light"
        android:onClick="doMul"
        android:text="x"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

 <Button
        android:id="@+id/button"
        android:layout_width="103dp"
        android:layout_height="46dp"
```

```
        android:layout_marginStart="113dp" android:layout_marginTop="356dp"
        android:layout_marginEnd="206dp" android:layout_marginBottom="199dp"
        android:backgroundTint="@android:color/holo_red_light"
        android:onClick="doMod"
        android:text="%(mod)" app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.515" />

<Button
        android:id="@+id/pow" android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="113dp"
        android:layout_marginTop="292dp"
        android:layout_marginEnd="210dp"
        android:layout_marginBottom="263dp"
        android:backgroundTint="@android:color/holo_red_light"
        android:onClick="doPow"
        android:text="n1^n2" app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.507" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.calculator2;

import android.os.Bundle;

import com.google.android.material.snackbar.Snackbar;

import androidx.appcompat.app.AppCompatActivity;

import android.text.TextUtils;
import android.view.View;
import androidx.navigation.NavController;
import androidx.navigation.Navigation;
import androidx.navigation.ui.AppBarConfiguration;
import androidx.navigation.ui.NavigationUI;
import com.example.calculator2.databinding.ActivityMainBinding;
import android.view.Menu;
import android.view.MenuItem;
import android.os.Bundle;
import android.view.View;
```

```
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private AppBarConfiguration appBarConfiguration;
    private ActivityMainBinding binding;
    public EditText e1, e2;
    TextView t1;
    int num1, num2;

    public boolean getNumbers() {

        e1 = (EditText) findViewById(R.id.num1);

        e2 = (EditText) findViewById(R.id.num2);

        t1 = (TextView) findViewById(R.id.result);

                String s1 = e1.getText().toString();

                String s2 = e2.getText().toString();

         if(s1.equals("Please enter value 1") && s2.equals(null))
         {
             String result = "Please enter value 2";
             e2.setText(result);
             return false;
         }
        if(s1.equals(null) && s2.equals("Please enter value 2"))
        {
            String result = "Please enter value 1";
            e1.setText(result);
            return false;
        }
        if(s1.equals("Please enter value 1") || s2.equals("Please enter value 2"))
        {
            return false;
        }

         if((!s1.equals(null) && s2.equals(null))|| (!s1.equals("") && s2.equals("")) ){

            String result = "Please enter value 2";

            e2.setText(result);
            return false;
        }
```

```
    if((s1.equals(null) && !s2.equals(null))|| (s1.equals("") && !s2.equals("")) ){
        String result = "Please enter value 1";
        e1.setText(result);
        return false;
    }
    if((s1.equals(null) && s2.equals(null))|| (s1.equals("") && s2.equals("")) ){
        String result1 = "Please enter value 1";
        e1.setText(result1);
        String result2 = "Please enter value 2";
        e2.setText(result2);
        return false;
    }

    else {
        num1 = Integer.parseInt(s1);
         num2 = Integer.parseInt(s2);

    }

    return true;
}

public void doSum(View v) {

            if (getNumbers()) {
        int sum = num1 + num2;
        t1.setText(Integer.toString(sum));
    }
   else
    {
        t1.setText("Error Please enter Required Values");
    }

}
public void clearTextNum1(View v) {
   e1.getText().clear();
}
public void clearTextNum2(View v) {
```

```
    e2.getText().clear();
}
public void doPow(View v) {

        if (getNumbers()) {
       double sum = Math.pow(num1, num2);
       t1.setText(Double.toString(sum));
   }
   else
   {
       t1.setText("Error Please enter Required Values");
   }
}

public void doSub(View v) {

   if (getNumbers()) {
       int sum = num1 - num2;
       t1.setText(Integer.toString(sum));
   }
   else
   {
       t1.setText("Error Please enter Required Values");
   }
}

   public void doMul(View v) {
   if (getNumbers()) {
       int sum = num1 * num2;
       t1.setText(Integer.toString(sum));
   }
   else
   {
       t1.setText("Error Please enter Required Values");
   }
}

public void doDiv(View v) {
   if (getNumbers()) {

       double sum = num1 / (num2 * 1.0);
       t1.setText(Double.toString(sum));
   }
   else
   {
       t1.setText("Error Please enter Required Values");
   }
}
```

```
    public void doMod(View v) {
        if (getNumbers()) {
            double sum = num1 % num2;
            t1.setText(Double.toString(sum));
        }
        else
        {
            t1.setText("Error Please enter Required Values");
        }
    }


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        e1 = (EditText) findViewById(R.id.num1);
        e2 = (EditText) findViewById(R.id.num2);  }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }

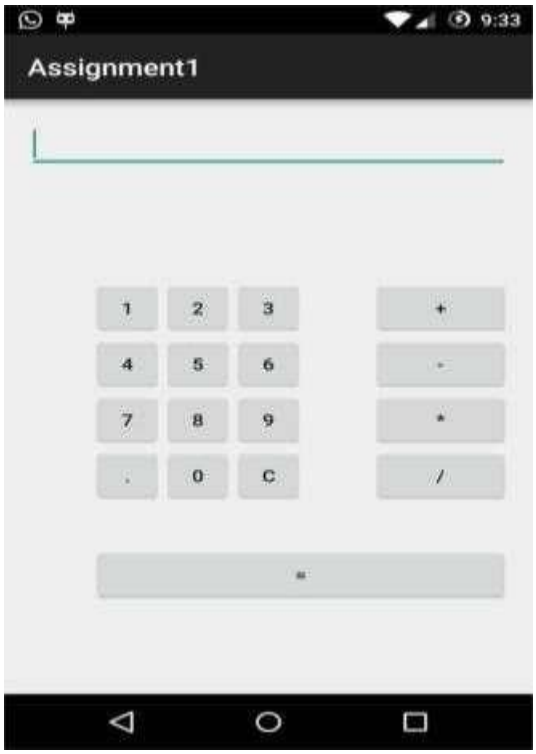
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.action_settings) {
            return true;
        }

        return super.onOptionsItemSelected(item);
    }

    @Override
    public boolean onSupportNavigateUp() {
        NavController navController = Navigation.findNavController(this,
R.id.nav_host_fragment_content_main);
        return NavigationUI.navigateUp(navController, appBarConfiguration)
                || super.onSupportNavigateUp();
    }
}
```

Output:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, a calculator mobile android application is developed and executed successfully.

EX.NO : 04	Android application for display the current location of the user
DATE :	

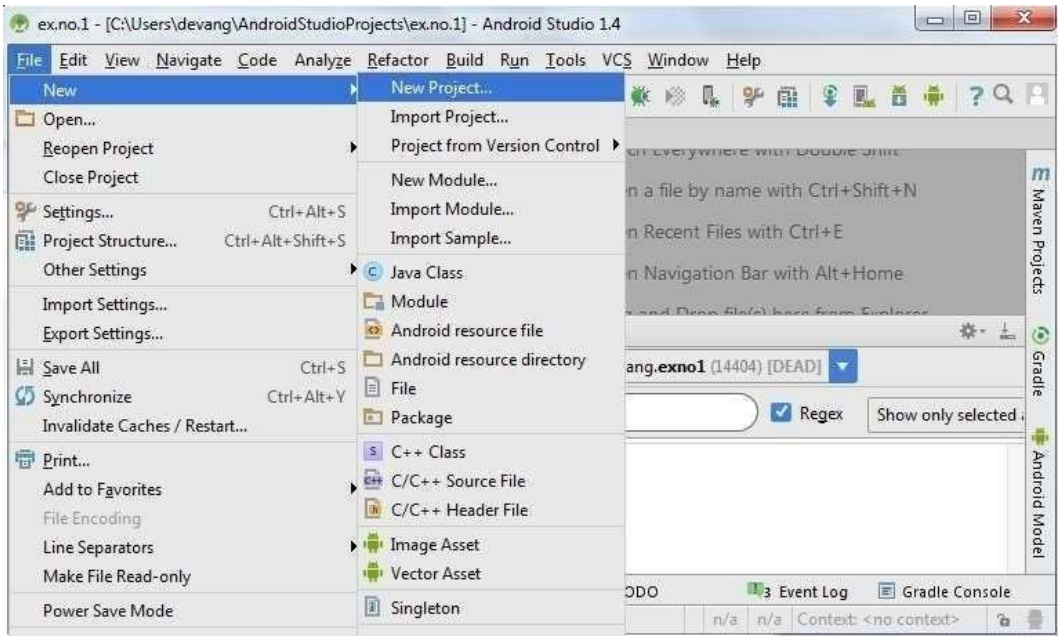
Aim:

To develop a simple android application that display the current location of the user .

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



- Click on Start a New Android Studio Project.
- Write your Application Name **"Current location"** and leave other fields blank as it is, then click NEXT.
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**. After selectingyour API click on NEXT.
- Select the **empty activity** and click NEXT.

- Leave the activity name "**MainActivity"** and leave everything as it is. Click F*inish*

Program

Acquiring Permissions in AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.getuserlocation">
```

```
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_BACKGROUND_LOCATION" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.GetUserLocation">
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Step 2. Designing the layout

```
?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#4caf50"
    android:gravity="center"
    android:orientation="vertical">

    <ImageView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:paddingBottom="120dp"
        android:src="@drawable/gfgimage" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:fontFamily="sans-serif-black"
        android:text="Latitude:" />

    <TextView
        android:id="@+id/latTextView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Latitude will be here! "
```

```
        android:textColor="#f5f5f5" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:fontFamily="sans-serif-black"
        android:text="Longitude:" />

    <TextView
        android:id="@+id/lonTextView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Longitude will be here! "
        android:textColor="#f5f5f5" />

</LinearLayout>
```

Step 3:MainActivity.java

```
import android.Manifest;
import android.annotation.SuppressLint;
import android.content.Context;
import  android.content.Intent;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationManager;
import android.os.Bundle;
import  android.os.Looper;
import  android.provider.Settings;
import   android.widget.TextView;
import android.widget.Toast;

import  androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

import com.google.android.gms.location.FusedLocationProviderClient;
import com.google.android.gms.location.LocationCallback;
import com.google.android.gms.location.LocationRequest;
import com.google.android.gms.location.LocationResult;
import com.google.android.gms.location.LocationServices;
import com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;

public class MainActivity extends AppCompatActivity {

    TextView latitudeTextView, longitTextView;
    int PERMISSION_ID = 44;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

        latitudeTextView = findViewById(R.id.latTextView);
        longitTextView = findViewById(R.id.lonTextView);

        mFusedLocationClient = LocationServices.getFusedLocationProviderClient(this);

        getLastLocation();
    }

    @SuppressLint("MissingPermission")
    private void getLastLocation() {

        if (checkPermissions()) {

            if (isLocationEnabled()) {

                mFusedLocationClient.getLastLocation().addOnCompleteListener(new
OnCompleteListener<Location>() {
                    @Override
                    public void onComplete(@NonNull Task<Location> task) {
                        Location location = task.getResult();
                        if (location == null) {
                            requestNewLocationData();
                        } else {
                            latitudeTextView.setText(location.getLatitude() + "");
                            longitTextView.setText(location.getLongitude() + "");
                        }
                    }
                });
            } else {
                Toast.makeText(this, "Please turn on" + " your location...", Toast.LENGTH_LONG).show();
                Intent intent = new Intent(Settings.ACTION_LOCATION_SOURCE_SETTINGS);
                startActivity(intent);
            }
        } else {
            requestPermissions();
        }
    }

    @SuppressLint("MissingPermission")
    private void requestNewLocationData() {

        LocationRequest mLocationRequest = new LocationRequest();
        mLocationRequest.setPriority(LocationRequest.PRIORITY_HIGH_ACCURACY);
        mLocationRequest.setInterval(5);
        mLocationRequest.setFastestInterval(0);
        mLocationRequest.setNumUpdates(1);
         mFusedLocationClient = LocationServices.getFusedLocationProviderClient(this);
        mFusedLocationClient.requestLocationUpdates(mLocationRequest, mLocationCallback,
```

```
Looper.myLooper());
    }

    private LocationCallback mLocationCallback = new LocationCallback() {

        @Override
        public void onLocationResult(LocationResult locationResult) {
            Location mLastLocation = locationResult.getLastLocation();
            latitudeTextView.setText("Latitude: " + mLastLocation.getLatitude() + "");
            longitTextView.setText("Longitude: " + mLastLocation.getLongitude() + "");
        } };

    private boolean checkPermissions() {
        return ActivityCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_COARSE_LOCATION) == PackageManager.PERMISSION_GRANTED
&& ActivityCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION) ==
PackageManager.PERMISSION_GRANTED;

  ActivityCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_BACKGROUND_LOCATION) ==
PackageManager.PERMISSION_GRANTED    }

    private void requestPermissions() {
        ActivityCompat.requestPermissions(this, new String[]{
                Manifest.permission.ACCESS_COARSE_LOCATION,
                Manifest.permission.ACCESS_FINE_LOCATION}, PERMISSION_ID);    }

    private boolean isLocationEnabled() {
        LocationManager locationManager = (LocationManager)
getSystemService(Context.LOCATION_SERVICE);
        return locationManager.isProviderEnabled(LocationManager.GPS_PROVIDER) ||
locationManager.isProviderEnabled(LocationManager.NETWORK_PROVIDER);
    }
     @Override
    public void
    onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull int[]
grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);

        if (requestCode == PERMISSION_ID) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
                getLastLocation();    } } }
    @Override
    public void onResume()
     {
        super.onResume();
        if (checkPermissions())
         {
```

getLastLocation(); } }}

Output:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, a mobile android application for displaying the current location of the user is developed and executed successfully.

EX.NO : 05	Android application for User Registration and Login with Local Data Storage
DATE :	

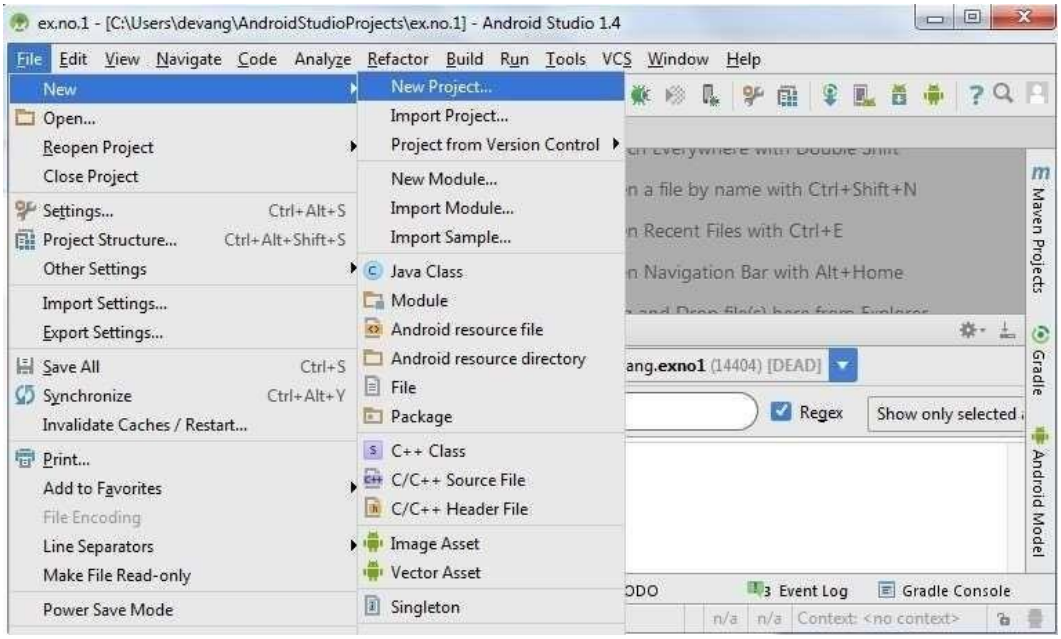
Aim:

Design and develop an android native application for user registration and login with local data storage

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



So lets gets started with our basic Android Simple Login App:

- Open your Android Studio.
- Click on Start a New Android Studio Project.
- Write your **Application Name :User registration** and leave other fields blank as it is
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**.
- Select the **empty activity** and click NEXT.

Install SQLite on Windows

- Step 1 − Go to SQLite download page, and download precompiled binaries from Windows section.
- Step 2 − Download sqlite-shell-win32-*.zip and sqlite-dll-win32-*.zip zipped files.
- Step 3 − Create a folder C:\>sqlite and unzip above two zipped files in this folder, which will give you sqlite3.def, sqlite3.dll and sqlite3.exe files.
- Step 4 − Add C:\>sqlite in your PATH environment variable and finally go to the command prompt and issue sqlite3 command, which should display the following result

Program : XML Layout for Registration

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".RegistrationActivity">

   <EditText
      android:id="@+id/usernameEditText"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Username"
      android:layout_marginTop="20dp"
      android:layout_marginHorizontal="16dp" />

   <EditText
      android:id="@+id/passwordEditText"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Password"
      android:inputType="textPassword"
      android:layout_below="@+id/usernameEditText"
      android:layout_marginTop="12dp"
      android:layout_marginHorizontal="16dp" />

   <Button
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Register"
      android:onClick="register"
      android:layout_below="@+id/passwordEditText"
      android:layout_marginTop="20dp"
      android:layout_marginHorizontal="16dp"  />    </RelativeLayout>
```

XML Layout for Login

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".LoginActivity">

   <EditText
      android:id="@+id/loginUsernameEditText"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Username"
```

```
    android:layout_marginTop="20dp"
    android:layout_marginHorizontal="16dp" />

  <EditText
    android:id="@+id/loginPasswordEditText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:layout_below="@+id/loginUsernameEditText"
    android:layout_marginTop="12dp"
    android:layout_marginHorizontal="16dp" />

  <Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"
    android:onClick="login"
    android:layout_below="@+id/loginPasswordEditText"
    android:layout_marginTop="20dp"
    android:layout_marginHorizontal="16dp" />   </RelativeLayout>
```

MainActivity.java

```
public class RegistrationActivity extends AppCompatActivity {
  EditText usernameEditText, passwordEditText;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_registration);
    usernameEditText = findViewById(R.id.usernameEditText);
    passwordEditText = findViewById(R.id.passwordEditText);      }

  public void register(View view) {
    String username = usernameEditText.getText().toString();
    String password = passwordEditText.getText().toString();      } }
```

LoginActivity.java

```
public class LoginActivity extends AppCompatActivity {
  EditText loginUsernameEditText, loginPasswordEditText;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_login);
    loginUsernameEditText = findViewById(R.id.loginUsernameEditText);
    loginPasswordEditText = findViewById(R.id.loginPasswordEditText);    }

  public void login(View view) {
    String username = loginUsernameEditText.getText().toString();
    String password = loginPasswordEditText.getText().toString();      }}
```

Output:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, a Android Application that demonstrate user registration and login with local data storage was executed and implemented successfully

EX.NO : 06	Android application for BMI calculator
DATE :	

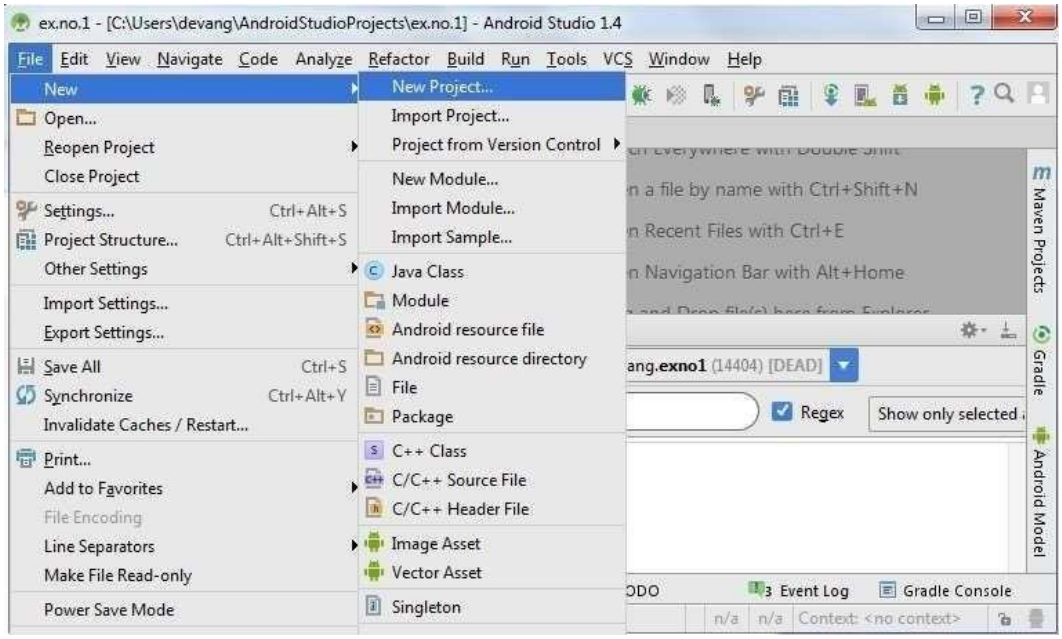
Aim:

To develop an android mobile application for a BMI calculator

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



So lets gets started with our basic Android Simple Login App:

- Open your Android Studio.
- Click on Start a New Android Studio Project.
- Write your **Application Name "BMI calculator"** and leave other fields blank as it is
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**.
- Select the **empty activity** and click NEXT.

Formula:

BMI = (weight) / (height * height

Program:

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp"
   tools:context=".MainActivity">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="BMI Calculator"
      android:textSize="24sp"
      android:layout_gravity="center_horizontal"
      android:layout_marginBottom="16dp"/>

   <EditText
      android:id="@+id/editTextWeight"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Weight (in kg)"
      android:inputType="numberDecimal"
      android:layout_marginBottom="16dp"/>

   <EditText
      android:id="@+id/editTextHeight"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Height (in meters)"
      android:inputType="numberDecimal"
      android:layout_marginBottom="16dp"/>

   <Button
      android:id="@+id/calculateButton"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Calculate BMI"
      android:layout_marginBottom="16dp"/>

   <TextView
      android:id="@+id/textViewResult"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center"
    android:textSize="18sp"/>    </LinearLayout>
```

Mainactivity.java

```
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

   private EditText editTextWeight;
   private EditText editTextHeight;
   private TextView textViewResult;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      editTextWeight = findViewById(R.id.editTextWeight);
      editTextHeight = findViewById(R.id.editTextHeight);
      textViewResult = findViewById(R.id.textViewResult);

      Button calculateButton = findViewById(R.id.calculateButton);
      calculateButton.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View v) {
            calculateBMI();              }        });      }

   private void calculateBMI() {
      String weightStr = editTextWeight.getText().toString();
      String heightStr = editTextHeight.getText().toString();

      if (!weightStr.isEmpty() && !heightStr.isEmpty()) {
         float weight = Float.parseFloat(weightStr);
         float height = Float.parseFloat(heightStr);
         float bmi = weight / (height * height);
         String result = "Your BMI: " + String.format("%.2f", bmi);
         textViewResult.setText(result);
      } else {
         textViewResult.setText("Please enter both weight and height.");          }     } }
```

OUTPUT :



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, a simple BMI android application was executed and implemented successfully.

EX.NO : 07	Android application for Expense manager
DATE :	

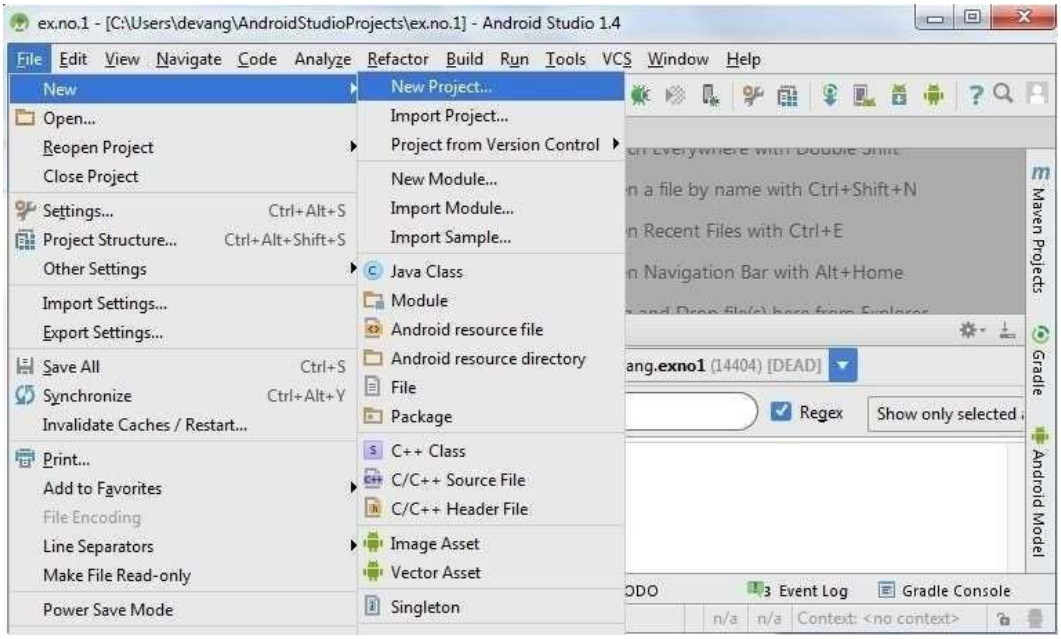
Aim:

Develop a simple expense manager which allows entering expenses and income on each day and displays category wise weekly income and expense

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



So lets gets started with our basic Android Simple Login App:

- Open your Android Studio.
- Click on Start a New Android Studio Project.
- Write your Application Name "**Expense manager**" and leave other fields blank
- Select the Minimum SDK "**API 15: Android 4.0.3(IceCreamSandwich)**".
- Select the empty activity and click NEXT.

Designing layout for the Android Application:

- Click on **app -> res -> layout -> activity_main.xml**.
- Now click on Text as shown below.
- Then delete the code which is there and type the code as given below.

Program:

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   tools:context=".MainActivity">

   <!-- User input section -->
   <EditText
      android:id="@+id/editTextExpense"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Expense"
      />

   <Button
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Add Expense"
      android:onClick="addExpense"
      />

   <EditText
      android:id="@+id/editTextIncome"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Income"
      />

   <Button
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Add Income"
      android:onClick="addIncome"
      />

   <!-- Weekly summary section -->
   <TextView
      android:id="@+id/textViewWeeklyExpense"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Weekly Expense: $0"
      />

   <TextView
```

```
        android:id="@+id/textViewWeeklyIncome"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Weekly Income: $0"
        />

</LinearLayout>
```

Mainactivity.java

```
import android.os.Bundle;
import android.support.v7.app.AppCompatActivity;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    private double weeklyExpense = 0.0;
    private double weeklyIncome = 0.0;

    private EditText editTextExpense;
    private EditText editTextIncome;
    private TextView textViewWeeklyExpense;
    private TextView textViewWeeklyIncome;

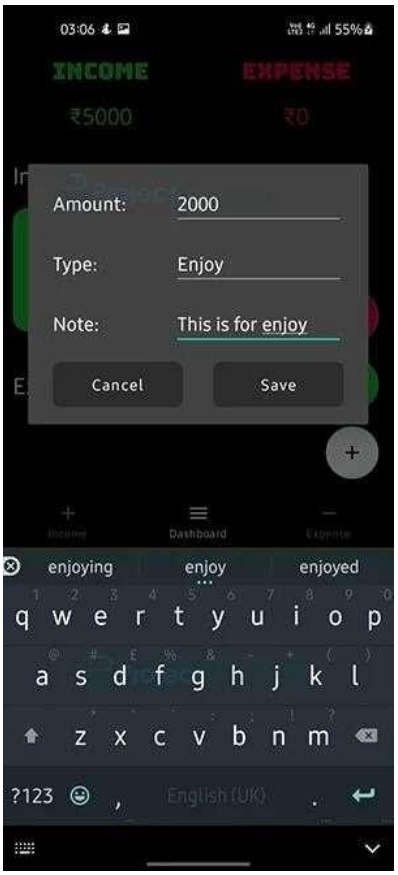
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextExpense = findViewById(R.id.editTextExpense);
        editTextIncome = findViewById(R.id.editTextIncome);
        textViewWeeklyExpense = findViewById(R.id.textViewWeeklyExpense);
        textViewWeeklyIncome = findViewById(R.id.textViewWeeklyIncome);
    }

    public void addExpense(View view) {
        double expense = Double.parseDouble(editTextExpense.getText().toString());
        weeklyExpense += expense;
        textViewWeeklyExpense.setText("Weekly Expense: $" + weeklyExpense);
        editTextExpense.setText("");
    }

    public void addIncome(View view) {
        double income = Double.parseDouble(editTextIncome.getText().toString());
        weeklyIncome += income;
        textViewWeeklyIncome.setText("Weekly Income: $" + weeklyIncome);
        editTextIncome.setText("");
    }
}
```

Output:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, an application that will have simple expense manager was developed and executed successfully.

EX.NO : 08	Android application for converting kilometers to miles
DATE :	

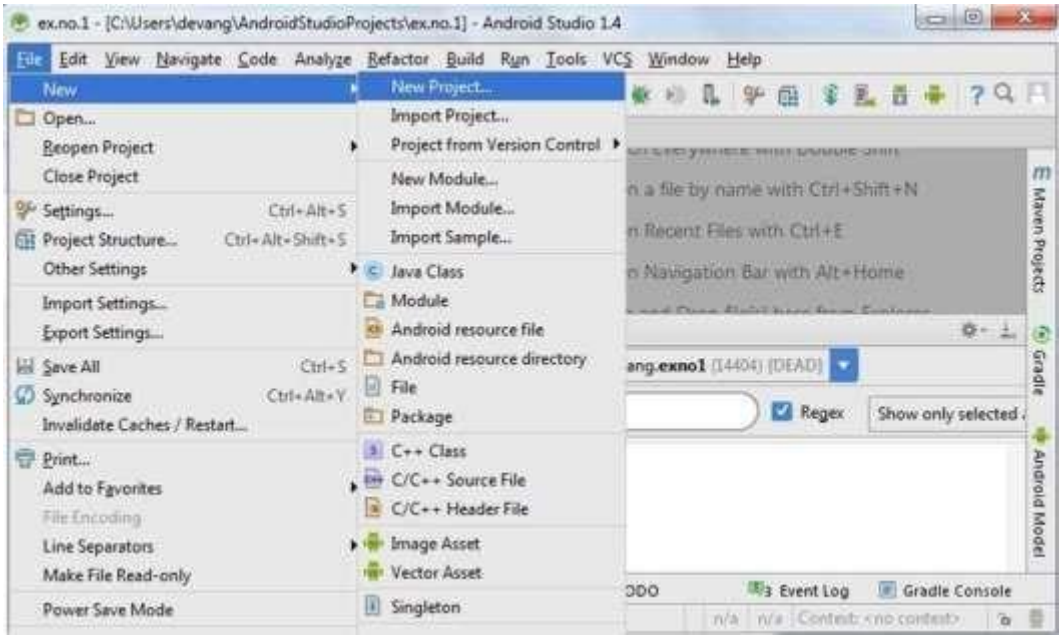
Aim:

To develop a mobile application to convert units from imperial system to metric system (km to miles, kg to pounds etc.,)

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



So lets gets started with our basic Android Simple Login App:

- Open your Android Studio.
- Click on Start a New Android Studio Project.
- Write your **Application Name "convert KM to Miles"** and leave other fields blank
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**.
- Select the **empty activity** and click NEXT.

Program:

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp"
   tools:context=".MainActivity">

   <EditText
      android:id="@+id/editTextValue"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter kilometers"
      android:inputType="numberDecimal"/>

   <Button
      android:id="@+id/buttonConvert"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Convert to Miles"/>

   <TextView
      android:id="@+id/textViewResult"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:layout_marginTop="16dp"
      android:text="Result will be shown here"
      android:textSize="18sp"/>
</LinearLayout>
```

MainActivity.java

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   private EditText editTextValue;
   private Button buttonConvert;
```

```
private TextView textViewResult;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

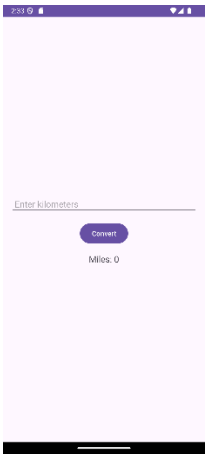
    editTextValue = findViewById(R.id.editTextValue);
    buttonConvert = findViewById(R.id.buttonConvert);
    textViewResult = findViewById(R.id.textViewResult);

    buttonConvert.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            convertToMiles();
        }
    });
}

private void convertToMiles() {
    try {
        double kilometers = Double.parseDouble(editTextValue.getText().toString());
        double miles = kilometers * 0.621371;

        textViewResult.setText(kilometers + " kilometers is equal to " + miles + " miles.");
    } catch (NumberFormatException e) {
        textViewResult.setText("Invalid input. Please enter a valid number.");
    }
}
```

Output:





Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, a simple Android app that converts kilometers to miles was developed and executed successfully.

EX.NO : 09	Android application for Student database in Cloud
DATE :	

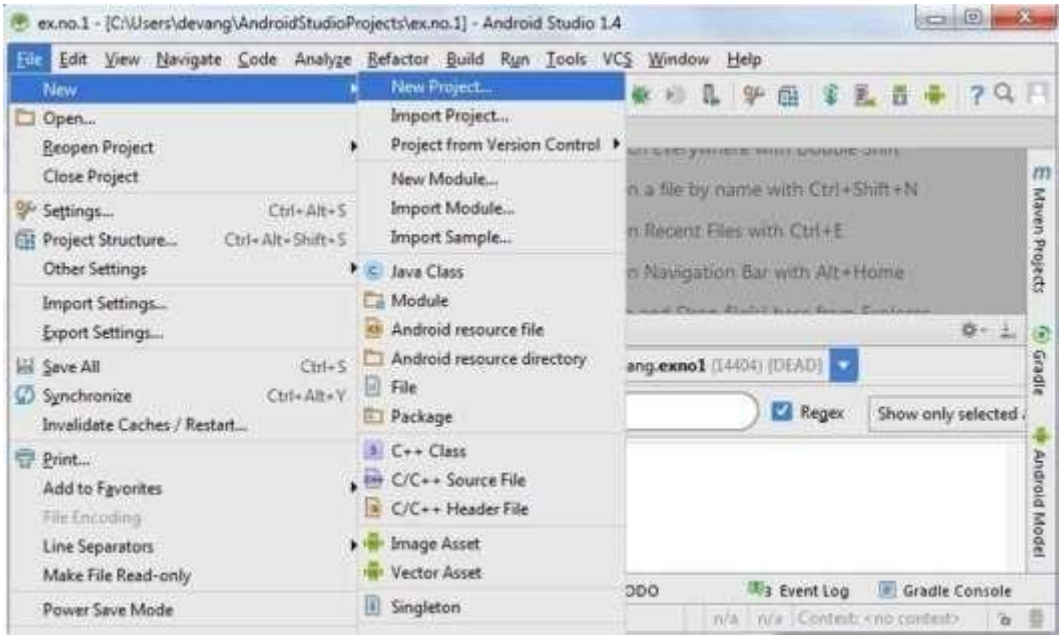
Aim:

To develop a mobile application to store and retrieve student details in a cloud database such as firebase.

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



So lets gets started with our basic Android Simple Login App:

- Open your Android Studio.
- Click on Start a New Android Studio Project.
- Write your **Application Name "Student database"** and leave other fields blank
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**.
- Select the **empty activity** and click NEXT.

Designing layout for the Android Application:

- Click on **app -> res -> layout -> activity_main.xml**.
- Now click on Text as shown below.

Program:

Activity_main.xml.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/editTextName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Student Name"/>

    <EditText
        android:id="@+id/editTextRollNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Roll Number"/>

    <Button
        android:id="@+id/buttonSave"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Save"/>

    <ListView
        android:id="@+id/listViewStudents"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"/>
</LinearLayout>
```

MainActivity.java

```
import android.os.Bundle;
```

```
import android.widget.ArrayAdapter;
import  android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;

public class MainActivity extends AppCompatActivity {

    private EditText editTextName, editTextRollNo;
    private Button buttonSave;
    private ListView listViewStudents;

    private DatabaseReference databaseReference;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextName = findViewById(R.id.editTextName);
        editTextRollNo = findViewById(R.id.editTextRollNo);
        buttonSave = findViewById(R.id.buttonSave);
        listViewStudents = findViewById(R.id.listViewStudents);

        databaseReference = FirebaseDatabase.getInstance().getReference("students");

        buttonSave.setOnClickListener(view -> saveStudent());


    }

    private void saveStudent() {
        String name = editTextName.getText().toString().trim();
        String rollNo = editTextRollNo.getText().toString().trim();

        if (!name.isEmpty() && !rollNo.isEmpty()) {
            String studentId = databaseReference.push().getKey();
            Student student = new Student(studentId, name, rollNo);

            databaseReference.child(studentId).setValue(student);

            editTextName.setText("");
            editTextRollNo.setText("");

            Toast.makeText(this, "Student saved", Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "Please fill in all fields", Toast.LENGTH_SHORT).show(); }    } }
```

Output:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, android application to store and retrieve student details in a firebase database is developed and executed successfully.

EX.NO : 10	Android application for day to day task (to-do) management
DATE :	

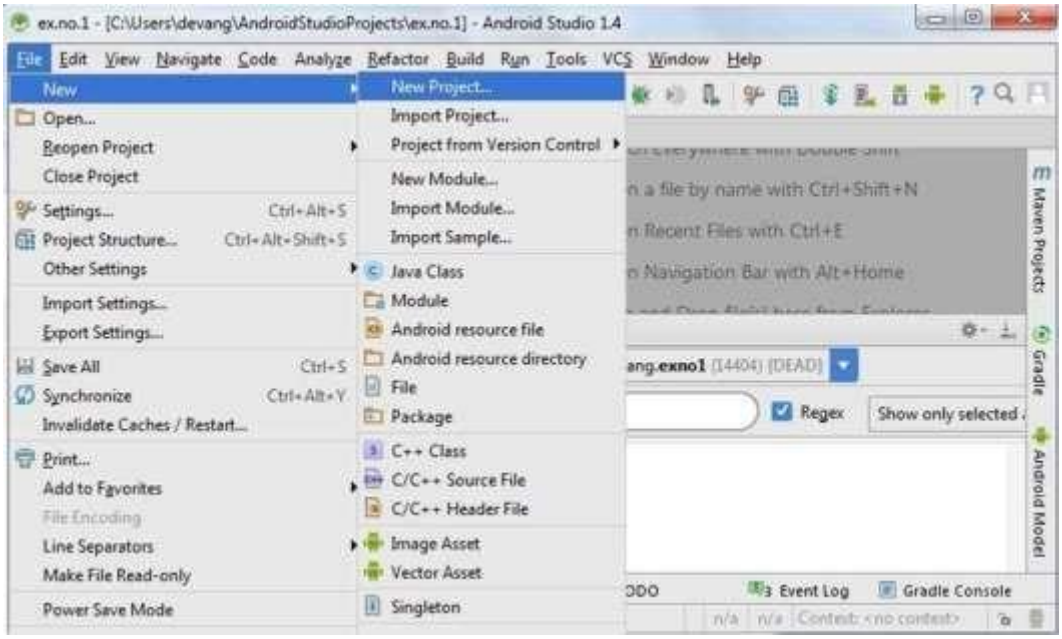
Aim:

To design and develop an application for day to day task (to-do) management

Procedure:

Creating a New project:

- Open Android Studio and then click on **File -> New -> New project**.



So lets gets started with our basic Android Simple Login App:

- Open your Android Studio.
- Click on Start a New Android Studio Project.
- Write your **Application Name "to-do management"** and leave other fields blank
- Select the **Minimum SDK "API 15: Android 4.0.3(IceCreamSandwich)"**.
- Select the **empty activity** and click NEXT.

.

Designing layout for the Android Application:

- Click on **app -> res -> layout -> activity_main.xml**.
- Now click on Text as shown below.

Program:

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ListView
        android:id="@+id/todoListView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_above="@+id/addTaskButton"  />

    <Button
        android:id="@+id/addTaskButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_alignParentBottom="true"
        android:text="Add Task" />

</RelativeLayout>
```

MainActivity.java

```
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import  android.widget.Button;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

   private ListView todoListView;
   private Button addTaskButton;
   private ArrayList<String> tasks;
   private ArrayAdapter<String> tasksAdapter;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
```

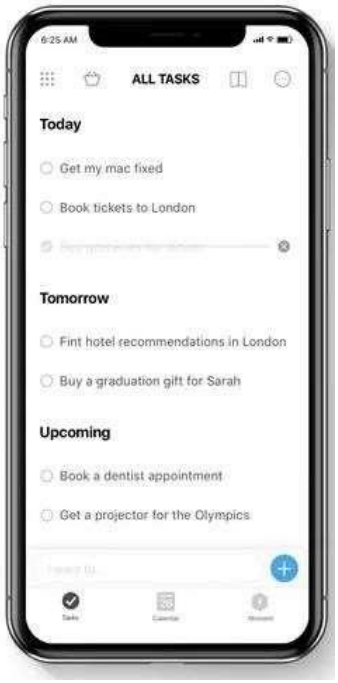
```
    setContentView(R.layout.activity_main);

    todoListView = findViewById(R.id.todoListView);
    addTaskButton = findViewById(R.id.addTaskButton);
    tasks = new ArrayList<>();
    tasksAdapter = new ArrayAdapter<>(this, android.R.layout.simple_list_item_1, tasks);
    todoListView.setAdapter(tasksAdapter);

    addTaskButton.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            String newTask = "New Task";
            tasks.add(newTask);
            tasksAdapter.notifyDataSetChanged();
        }
    });
  }
}
```

Output:



Department of CSE		
Aim	4	
Algorithm	4	
Program	4	
Output/Result	4	
Viva	4	
Total	20	

Result:

Thus, android application that uses native day to day task (to-do) management applicationis developed and executed successfully.